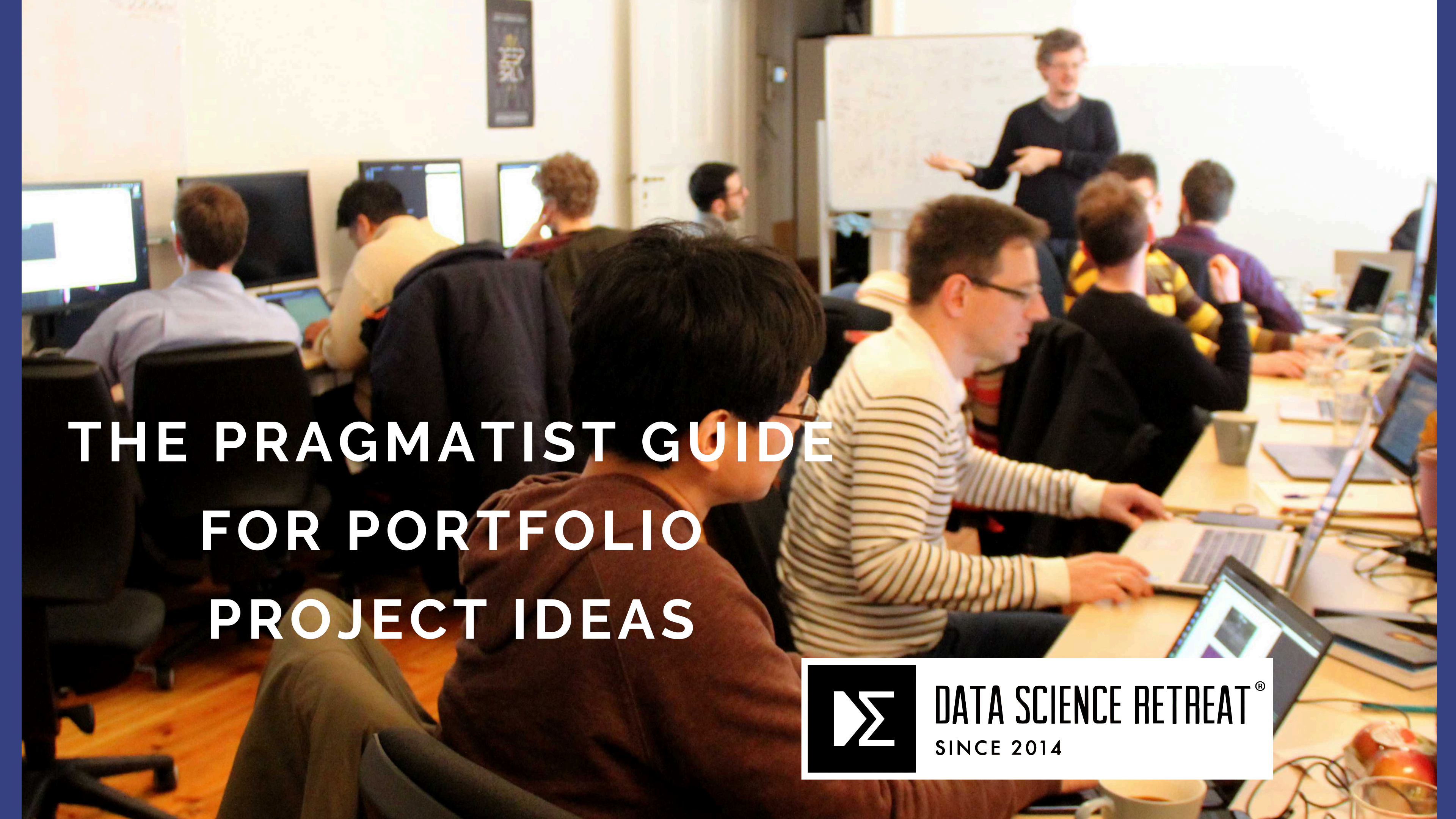
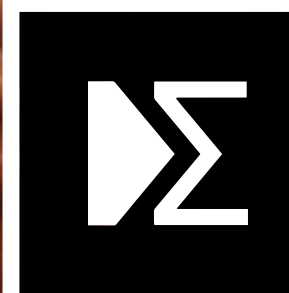




THE PRAGMATIST GUIDE FOR PORTFOLIO PROJECT IDEAS



DATA SCIENCE RETREAT®
SINCE 2014

Why I'm a relevant source of advice

- **Thinking/decision making** Did research in 3 out of the 5 top labs (CMU, Warwick -Chater-, Max Planck -Gigerenzer-)
- **Machine learning** (PhD UC Boulder, Postdoc CMU. 24 years experience, >230 projects)
- **Business** running Data Science Retreat for close to 10 years. Machine learning school, some alumni are at C-level, others have started companies with their portfolio project

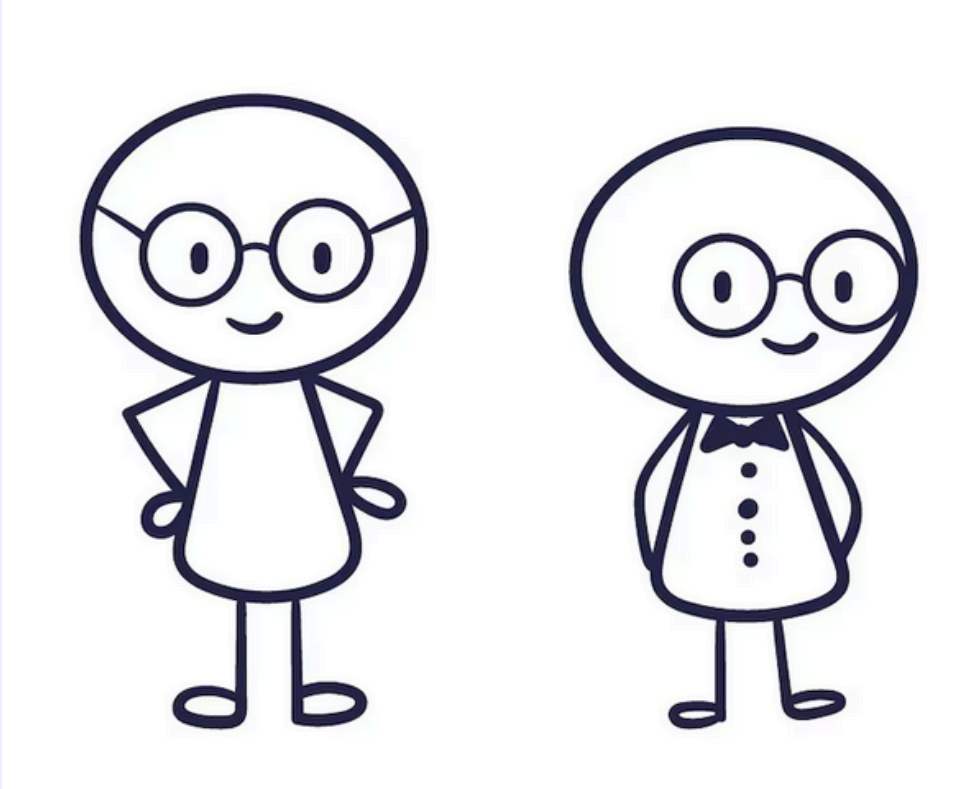
Launched in Berlin 2014, SF 2018, Toronto 2019. Self-funded (on a shoestring), bootstrapped, solo.



Problem solving vs problem finding

EXPECTATIONS

**EXPECTATIONS
EVERYWHERE**



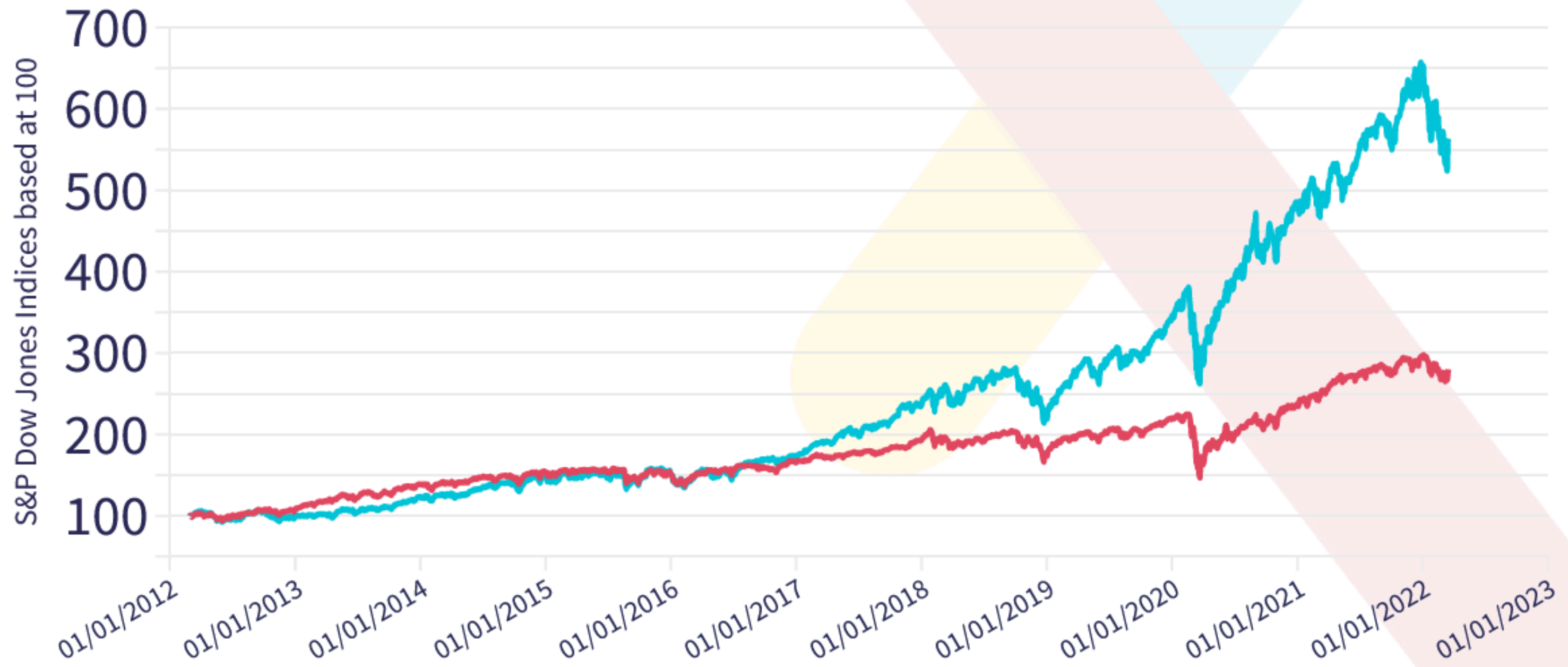
90%
OF STARTUPS
FAIL

Source: startup genome

Performance of Tech Companies and Non-Tech Companies

Tech Companies Grew 2.3X More Since the Pandemic Than Non-Tech Companies

■ S&P 500 Information Technology ■ S&P 500 Excluding Information Technology



Ecosystem by the Numbers

ECOSYSTEM VALUE ⓘ
(H2 2020 - 2022)

\$100 bn



GLOBAL AVG. \$34.6 bn

NUMBER OF UNICORNS ⓘ
(H2 2020 - 2022)

17



GLOBAL AVG. 4

SOFT. ENGINEER SALARY ⓘ
(2022)

\$56 k



GLOBAL AVG. \$46 k

TIME TO EXIT ⓘ
(2018 - 2022)

7.3 years



GLOBAL AVG. 9 years

TOTAL EARLY-STAGE FUNDING ⓘ
(H2 2020 - 2022)

\$5.8 bn



GLOBAL AVG. \$970 m

MEDIAN SEED ROUND ⓘ
(H2 2020 - 2022)

\$1.5 m



GLOBAL AVG. \$821 k

EARLY-STAGE FUNDING GROWTH ⓘ
(2019 - 20 VS. 2021 - 22)



Scale of 1 (lowest) to 10 (highest)

EXIT AMOUNT ⓘ
(2018 - 2022)

\$18 bn



GLOBAL AVG. \$11.3 bn

ECOSYSTEM VALUE GROWTH ⓘ
(H2 2020 - H2 2022 VS. H2 2018 - H2 2020)

47%



GLOBAL AVG. 47%

MEDIAN SERIES A ROUND ⓘ
(H2 2020 - 2022)

\$11 m



GLOBAL AVG. \$6 m

TOTAL VC FUNDING ⓘ
(2018 - 2022)

\$31 bn



GLOBAL AVG. \$6.6 bn

EXIT COUNT (#) ⓘ
(2018 - 2022)

337

GLOBAL AVG. 91



2018 2019 2020 2021 2022

Pragmatism

AND PRACTICAL REASONING

Pragmatism is a philosophical tradition that considers words and thought as tools and instruments for prediction, problem solving, and action, and rejects the idea that the function of thought is to describe, represent, or mirror reality. Pragmatists contend that most philosophical topics—such as the nature of knowledge, language, concepts, meaning, belief, and science—are all best viewed in terms of their **practical uses** and successes.

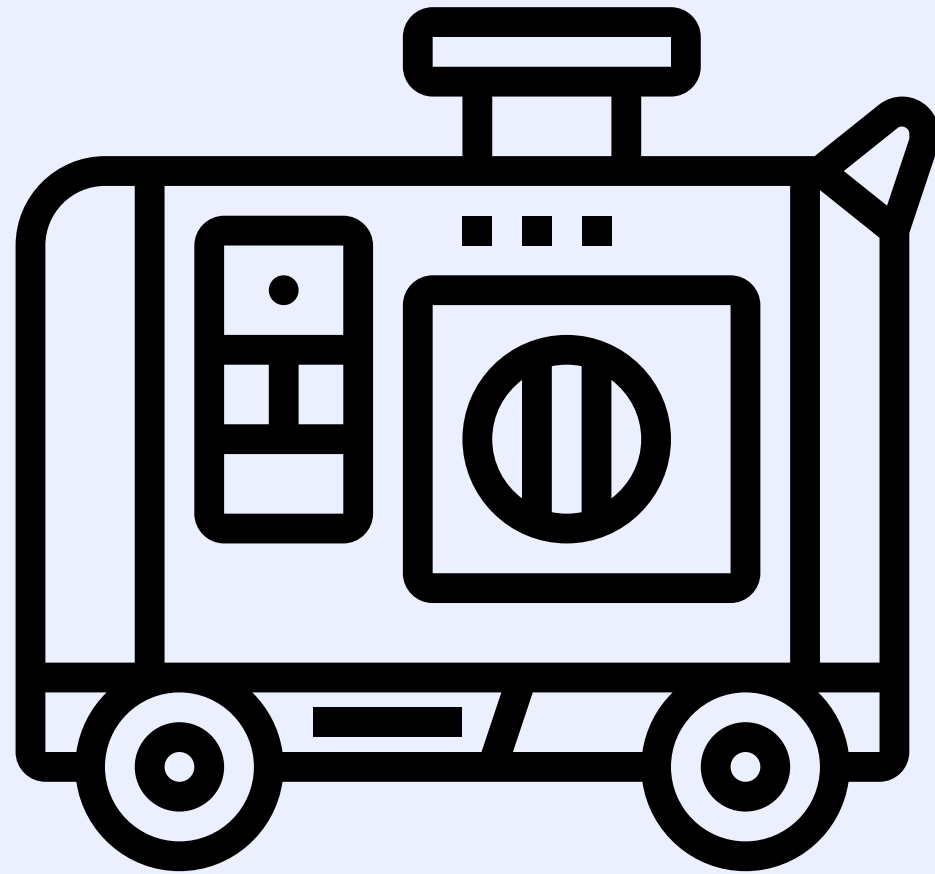
Goal

FIND A JOB

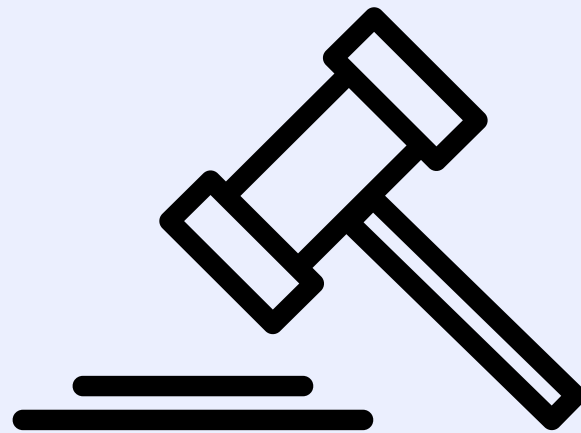
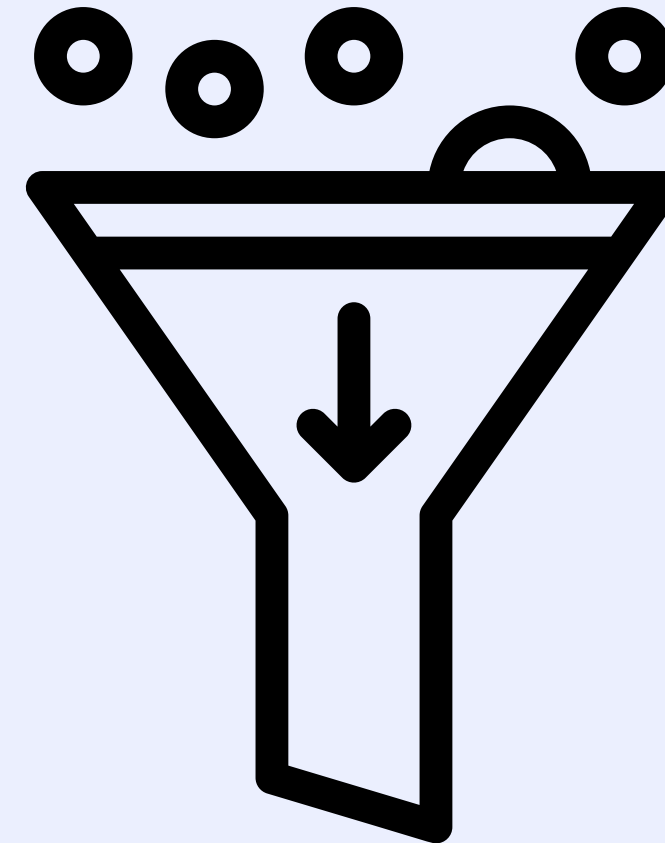
Constraints

- ML PROJECT
- 2 MONTHS TO FIND PROBLEM, WHILE TAKING CLASSES
- 1 MONTH TO IMPLEMENT AND PRESENT

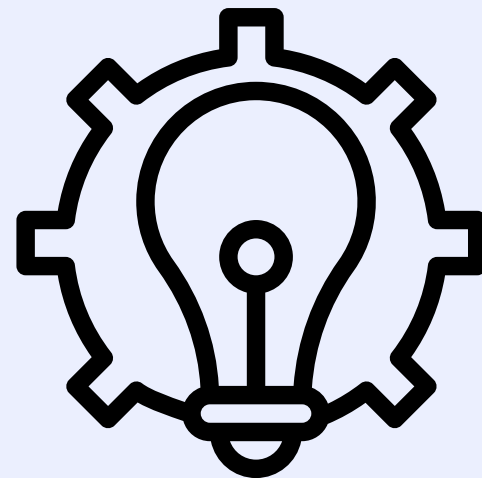
1. generation



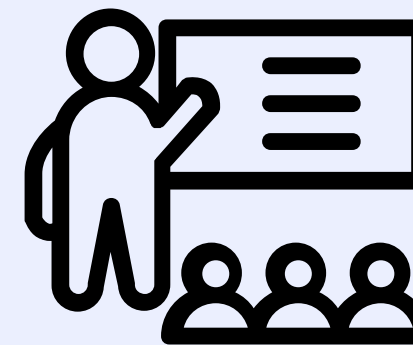
2. filtering



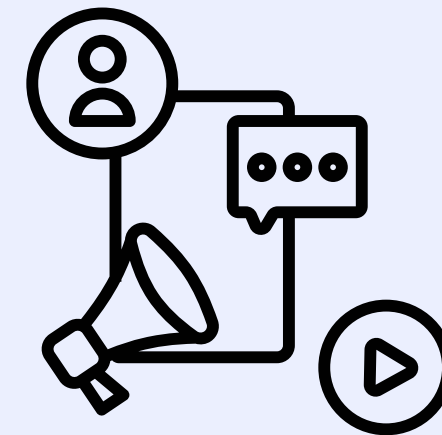
3. decision



4. implementation

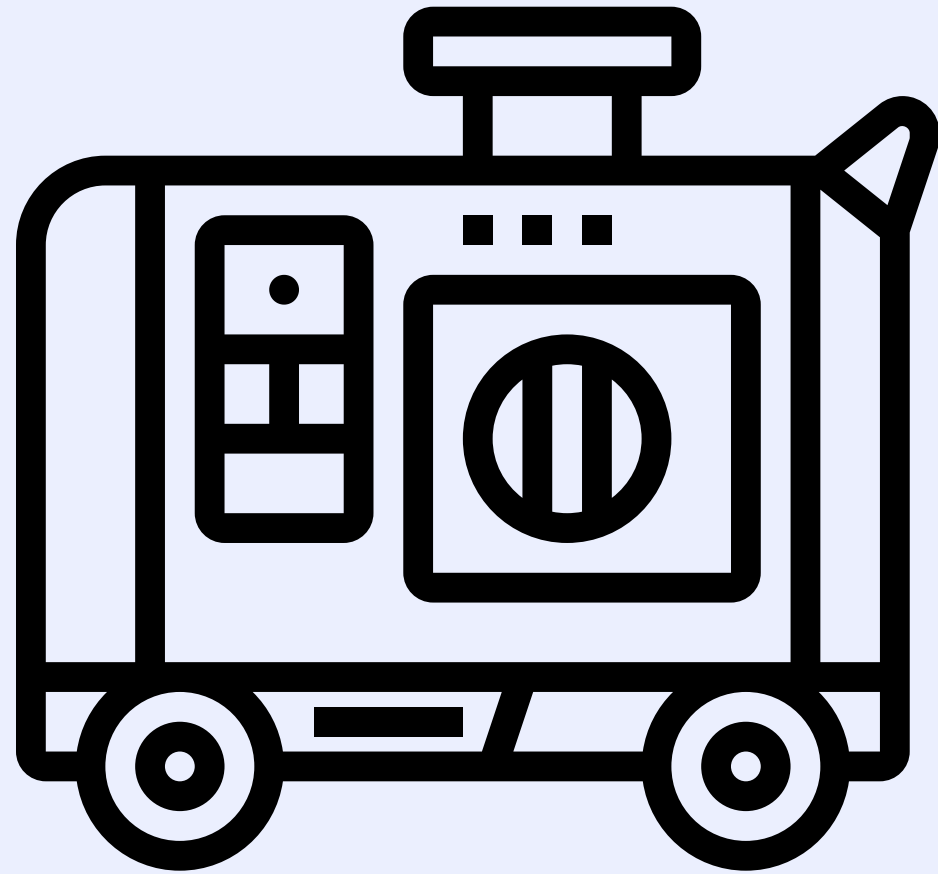


5. demo day



6. presence,
outreach

1. generation



- Problems you have (example, disphonia)
- Problems your previous organization has
- Problems so see solved with ML in a recent paper
- Problems solved with data you generate (sensors, hand labeling)
- Problems that when solved, can make or save money for an individual

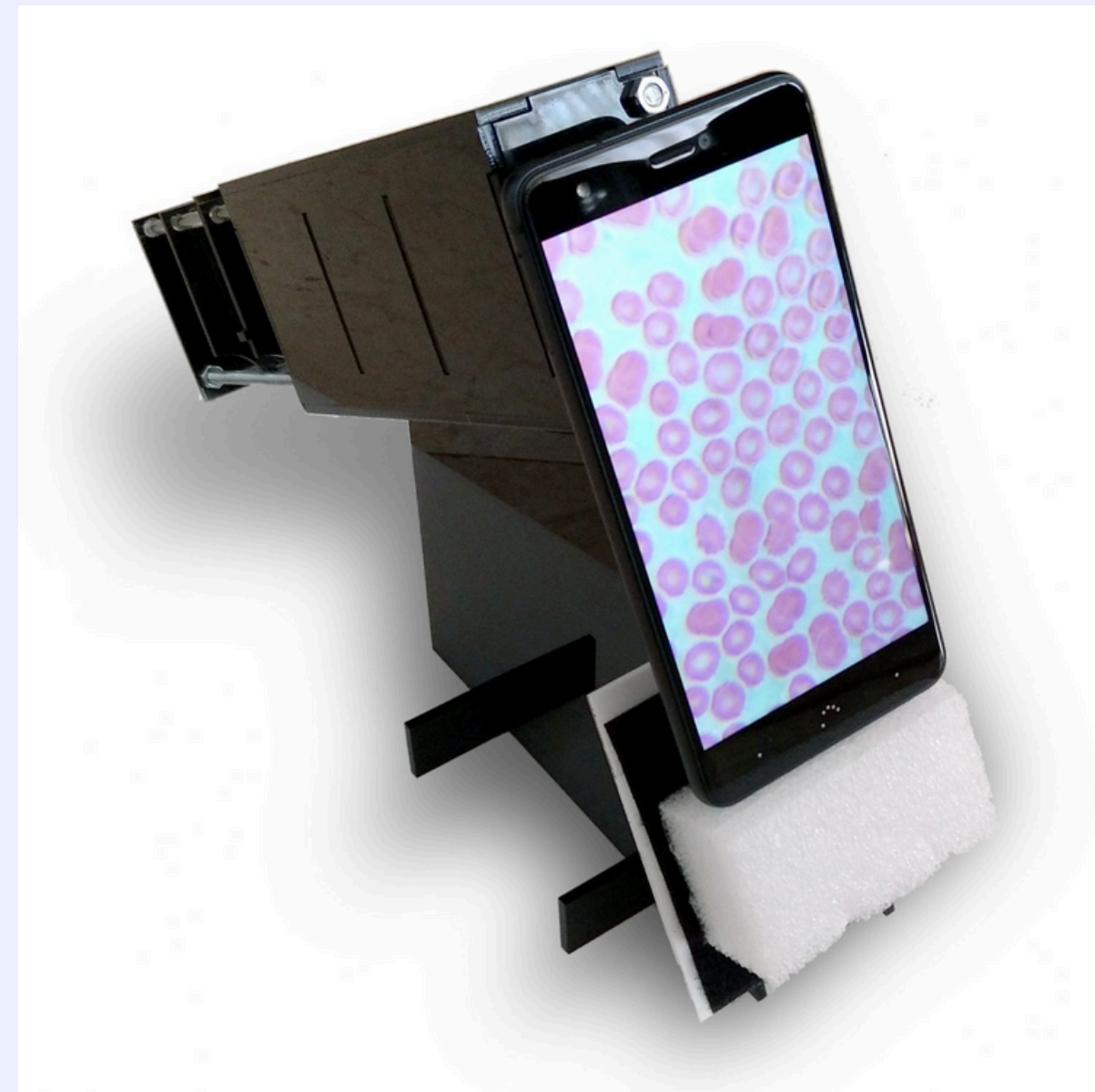
FLASHINESS

MARKETABILITY/
MEMORABILITY

EMPLOYABILITY

SOCIAL IMPACT

<https://aiscope.net/>

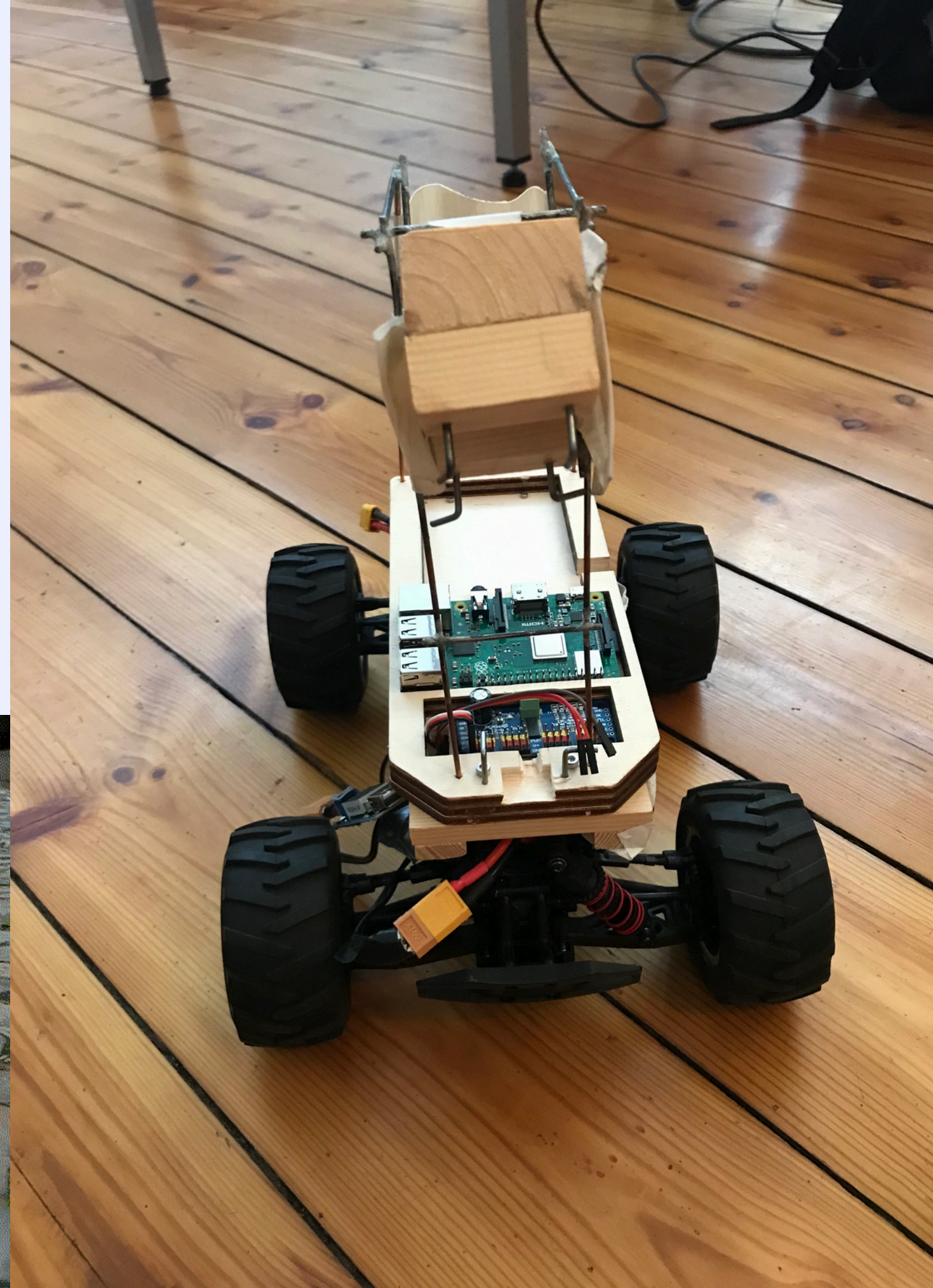
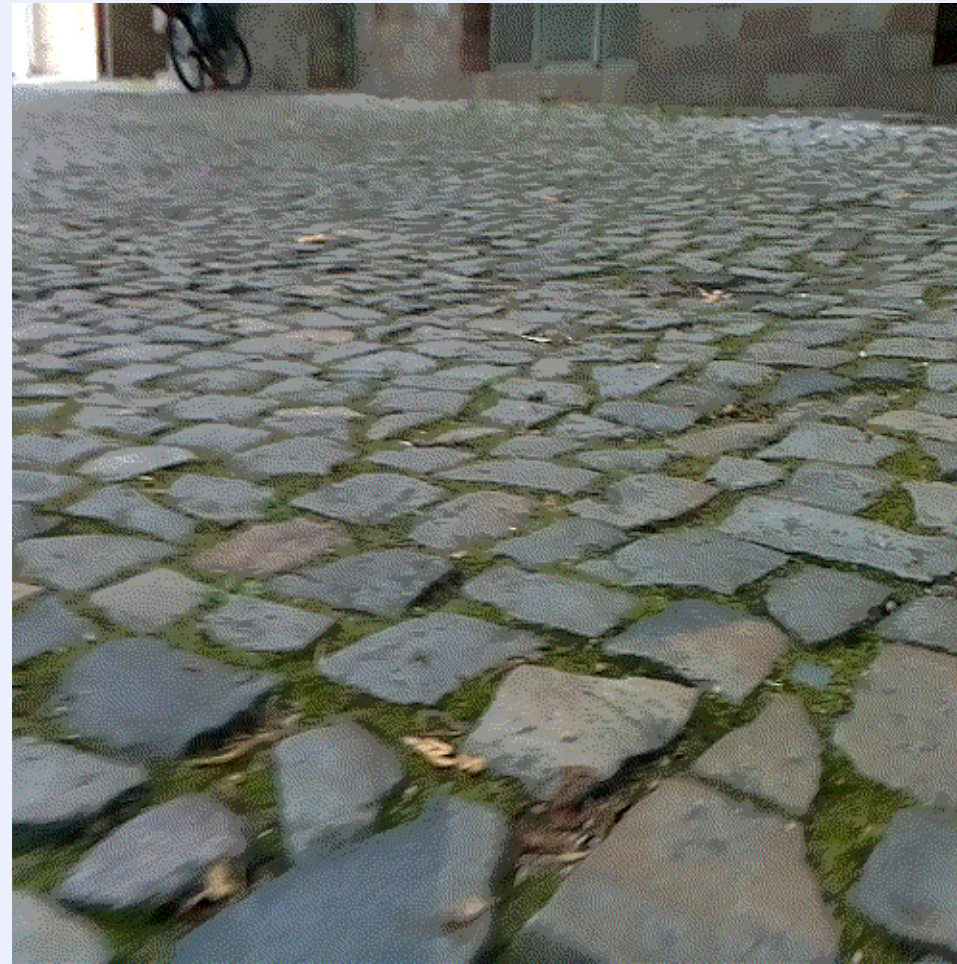


FLASHINESS

MARKETABILITY/
MEMORABILITY

HARDWARE

<https://emilypicksup.wordpress.com/>

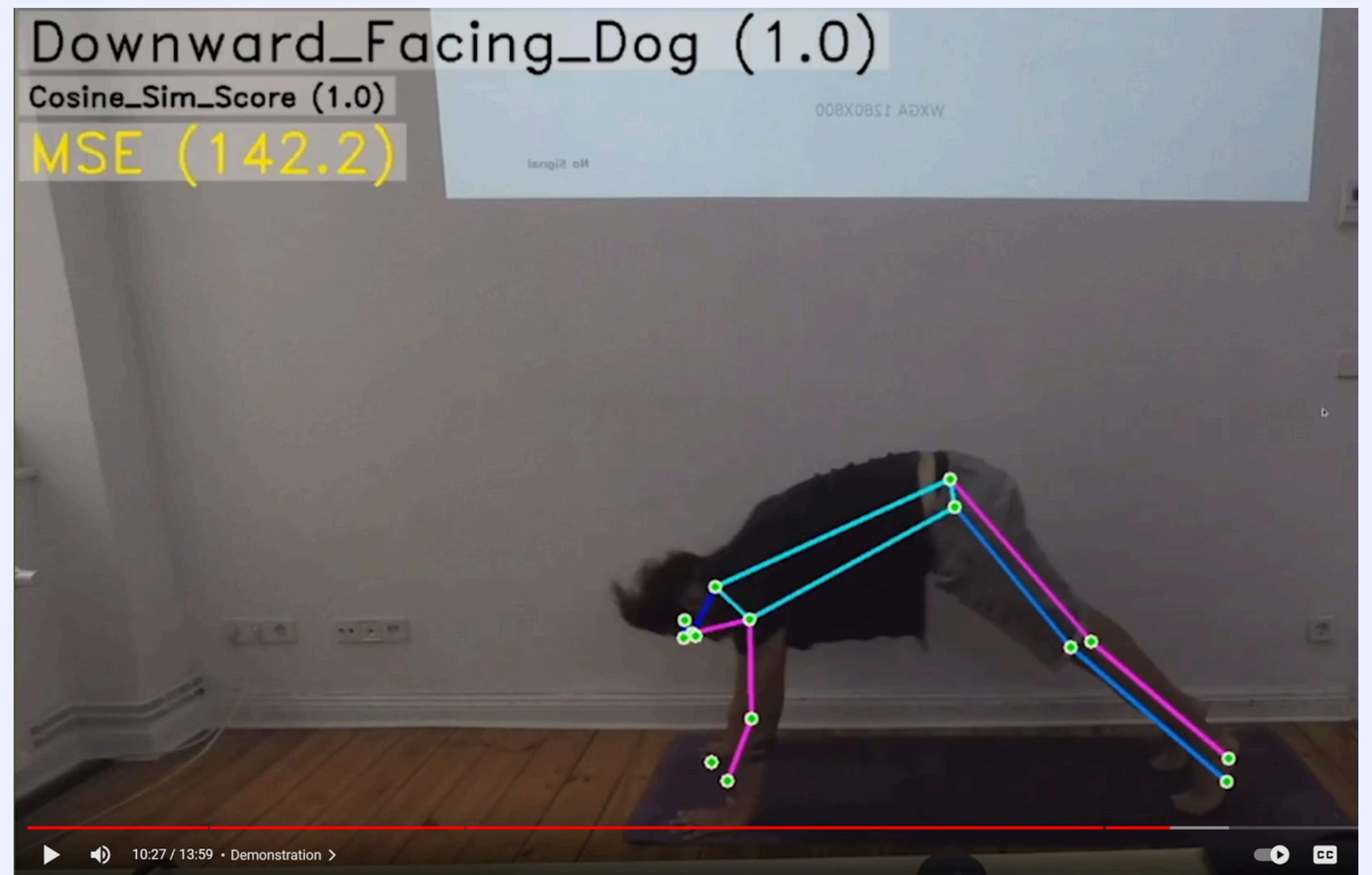


FLASHINESS

MARKETABILITY/
MEMORABILITY

EMPLOYABILITY

**REAL PROBLEM
THAT PRODUCES
BUSINESS VALUE**





ESTHETICS



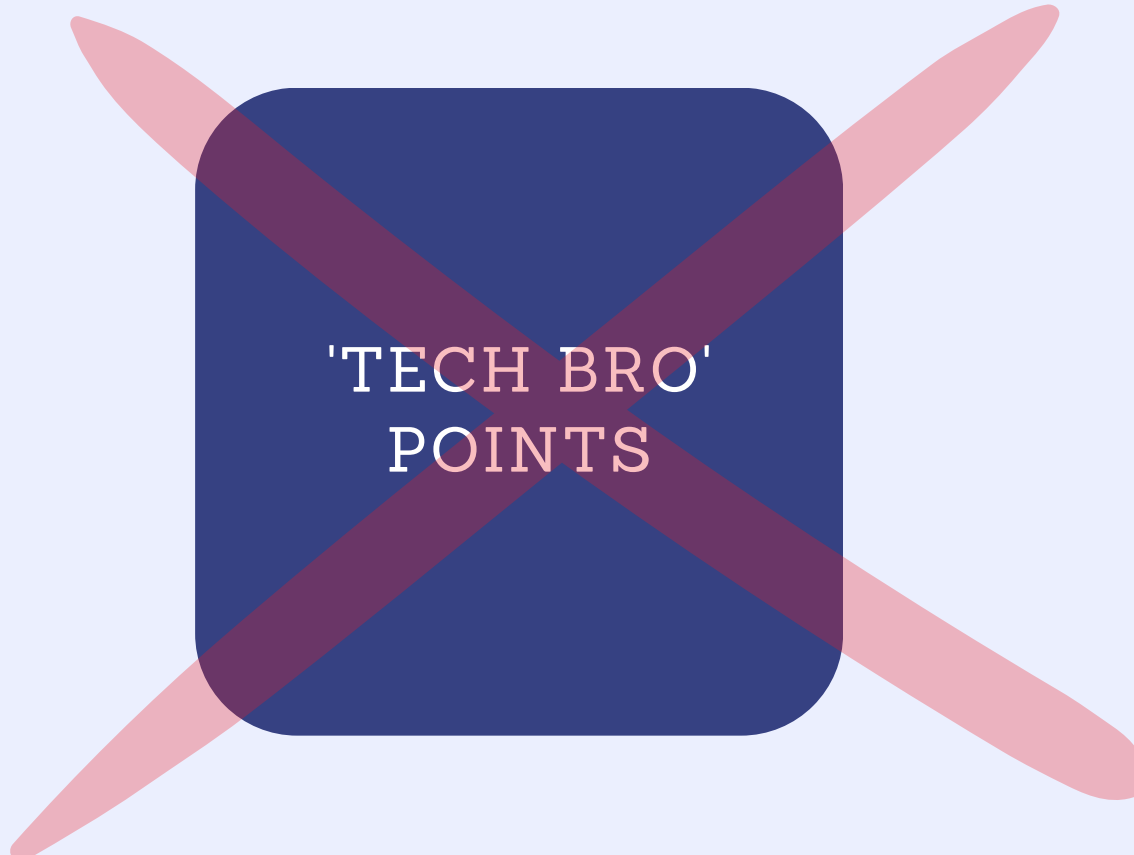
INTEREST, SELF-
PLEASURE



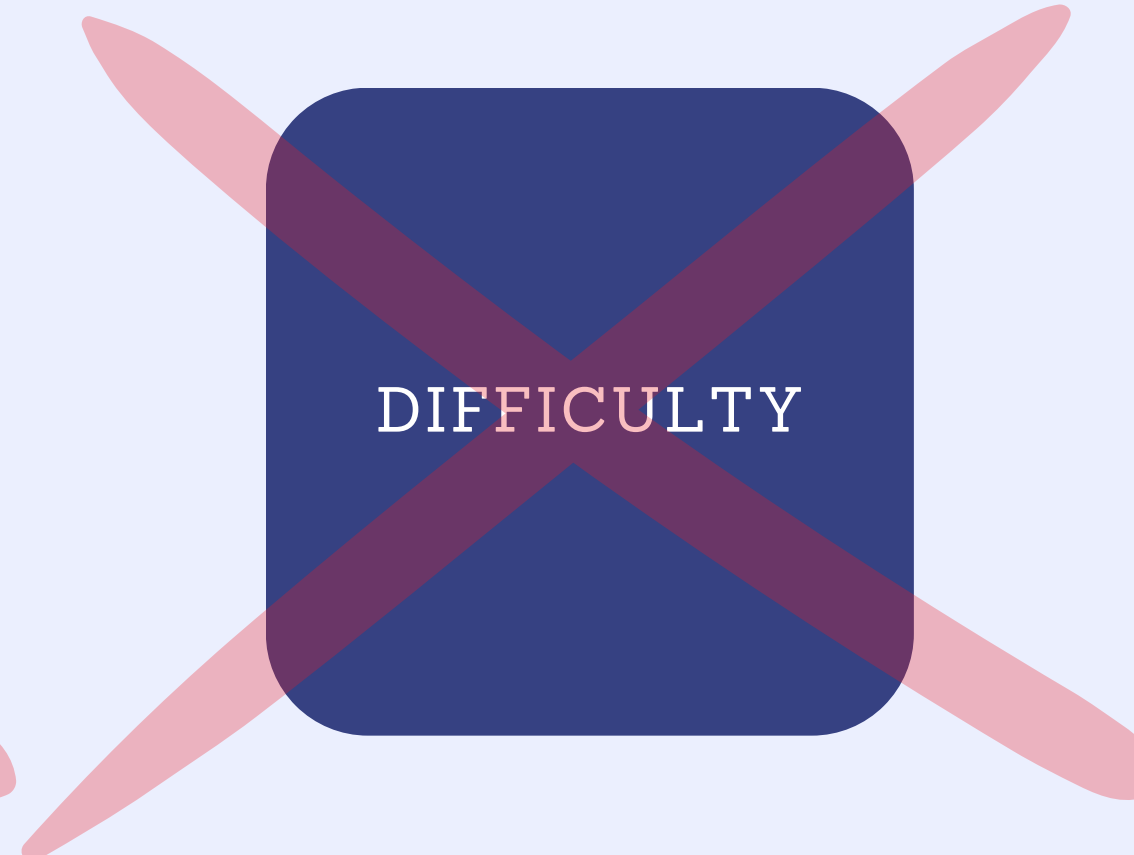
CONVENIENCE,
DATA
AVAILABLE



ORIGINALITY

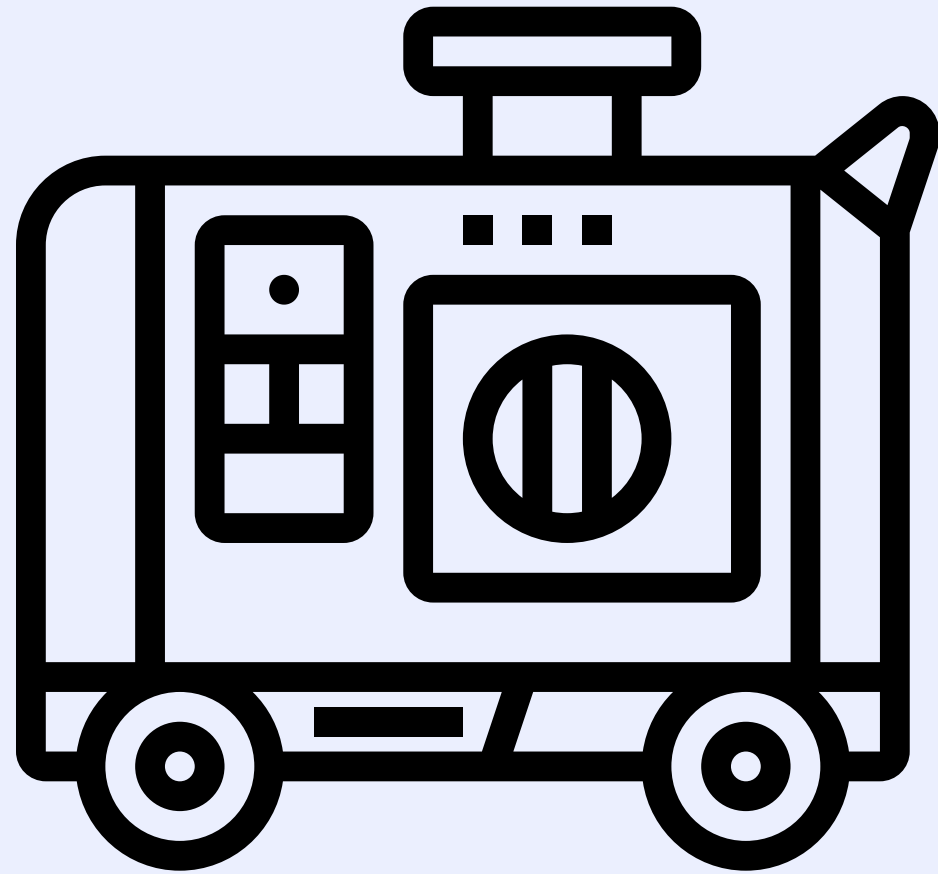


'TECH BRO'
POINTS



DIFFICULTY

1. generation



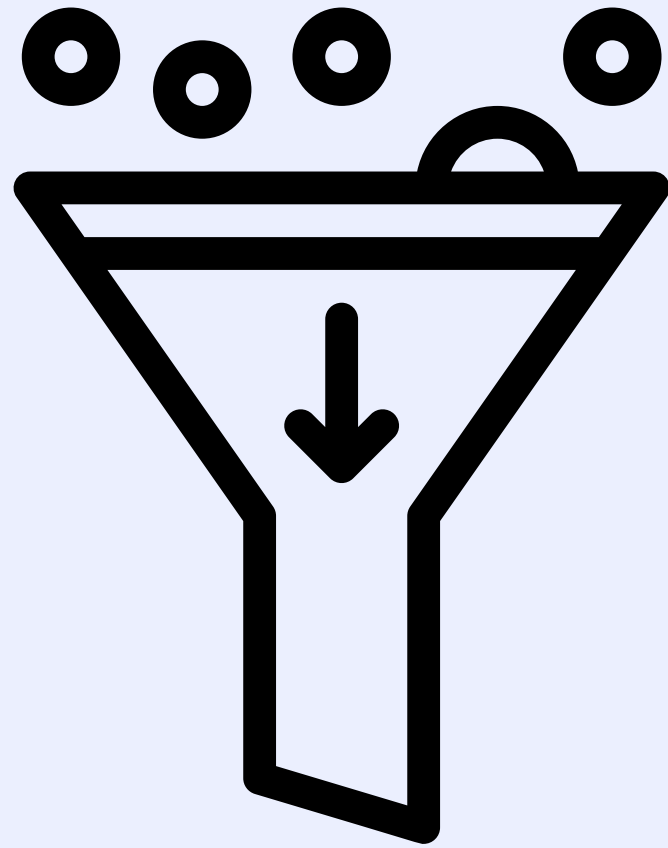
- Recurring problems in public forums (eg, r/berlin, hard to find a flat)
- Problems dictated by common sense, where value is obvious
- Problems validated by a previous attempt that failed (failed startups)
- Problems that become solvable just now by a recent development in ML
- Problems where you have domain knowledge that is hard to acquire

How to kill ideas

BEFORE INVESTING TIME AND MONEY IN THEM

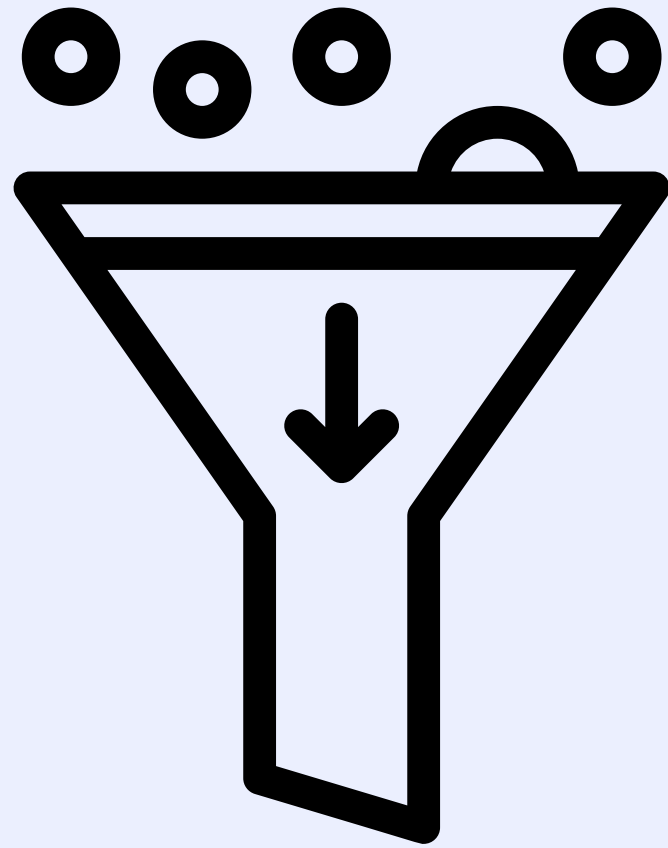
- THE PRAGMATIC WAY
- FROM FIRST PRINCIPLES/DEDUCTION
- FROM EMPIRICAL DATA/INDUCTION

2. filtering



- No 3rd party dependency
- There is a gold standard (supervised learning)
- Assumptions can be validated with logic/common sense rather than empirically
- Is the data available (or we can generate it in a reasonable time?)

2. filtering



- Problem is easy to communicate (easy to demo, visual, tells a story)
- Problem is a pain killer, not an aspirin
- You have direct access to your users, and they can give you feedback
- Problem is judged as doable (time-wise, complexity-wise) by someone with credible experience

THINKING FROM FIRST PRINCIPLES IN THE LLM ERA

(1) LLMs can elevate your productivity and learning speed

(2) But they lie

(3) So we humans need to elevate our thinking

(4) I present a method to evaluate epistemic value of texts that you can adapt to filtering ideas for portfolio projects

Debugging your thinking





this is madness!

The logic of real arguments

Second edition

A L E C F I S H E R

CAMBRIDGE

ARGUMENT ON THE VERACITY OF HOROSCOPES

I turned my attention to the case of twins, who are generally born within a short time of each other. Whatever significance in the natural order the astrologers may attribute to this interval of time, it is too short to be appreciated by human observation and no allowance can be made for it in the charts which an astrologer has to consult in order to cast a true horoscope. His predictions, then, will not be true, because he would have consulted the same charts for both Esau and Jacob and would have made the same predictions for each of them, whereas it is a fact that the same things did not happen to them both. Therefore, either he would have been wrong in his predictions or, if his forecast was correct, he would not have predicted the same future for each. And yet he would have consulted the same chart in each case. This proves that if he had foretold the truth, it would have been by luck, not by skill.

FROM ST AUGUSTINE'S CONFESSIONS

ARGUMENT ON THE VERACITY OF HOROSCOPES

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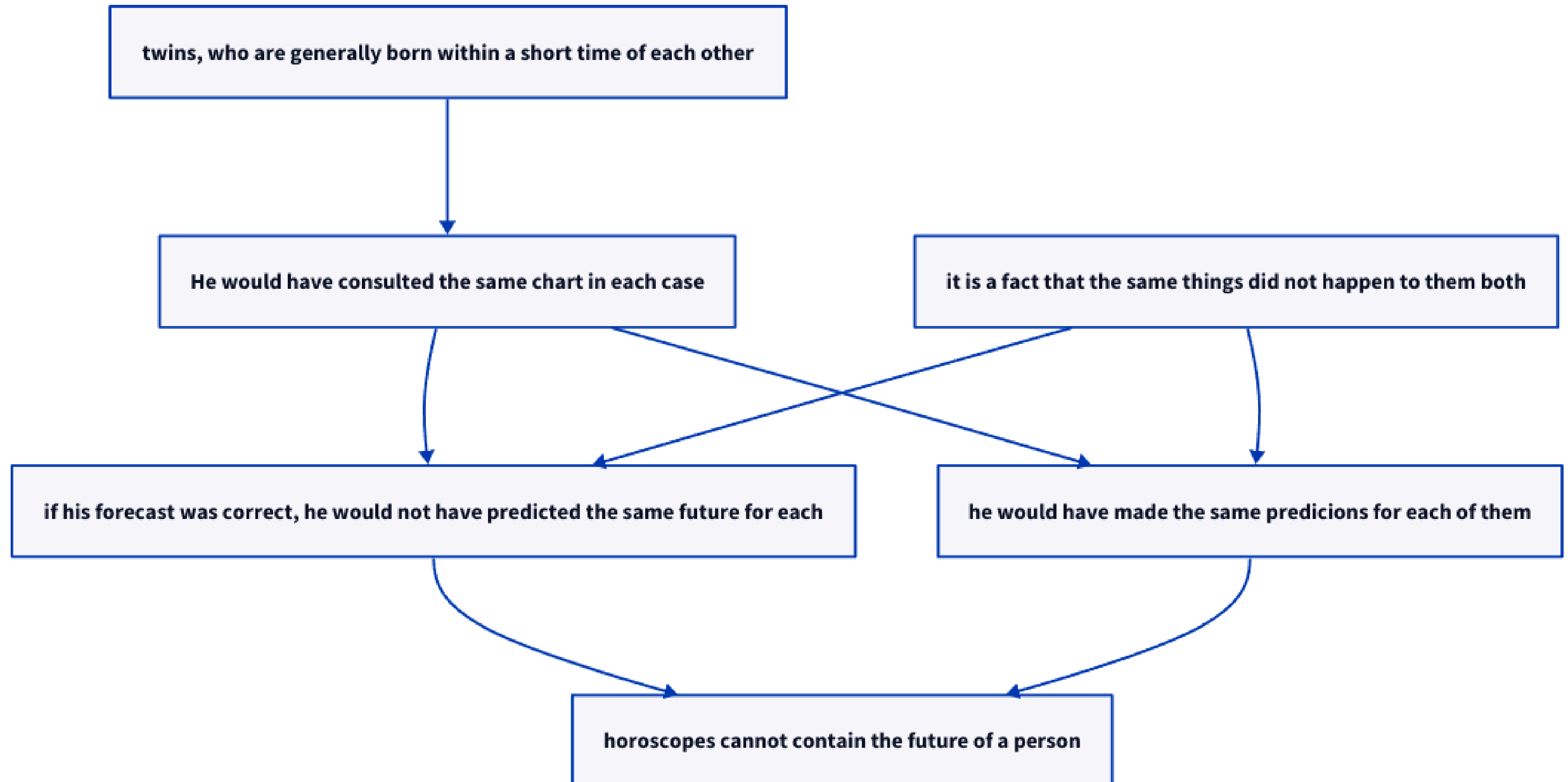
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it is a fact that the same things did not happen to them both

if his forecast was correct, he would not have predicted the same future for each

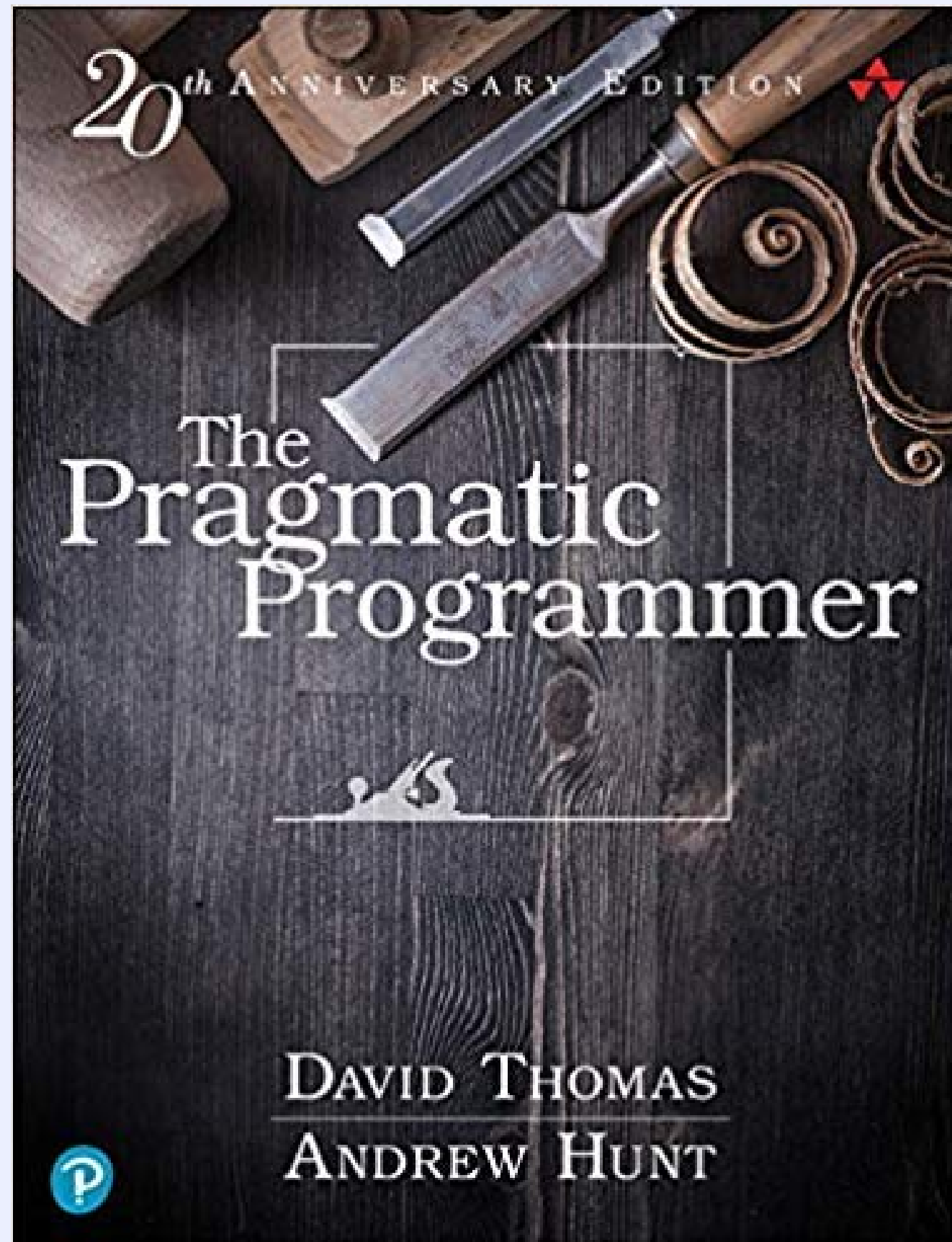
he would have made the same predictions for each of them

horoscopes cannot contain the future of a person



What argument or evidence would
justify me in believing the conclusion?





DECOUPLING CODE

In Topic 8, The Essence of Good Design we claim that using good design principles will make the code you write easy to change. Coupling is the enemy of change, because it links together things that must change in parallel. This makes change more difficult: either you spend time tracking down all the parts that need changing, or you spend time wondering why things broke when you changed “just one thing” and not the other things to which it was coupled

DOCUMENT

you spend time tracking down
all the parts that need
changing

DOCUMENT

you spend time wondering why
things broke when you
changed "just one thing" and
not the other things to which it
was coupled

DOCUMENT

[coupling] links together
things that must change in
parallel

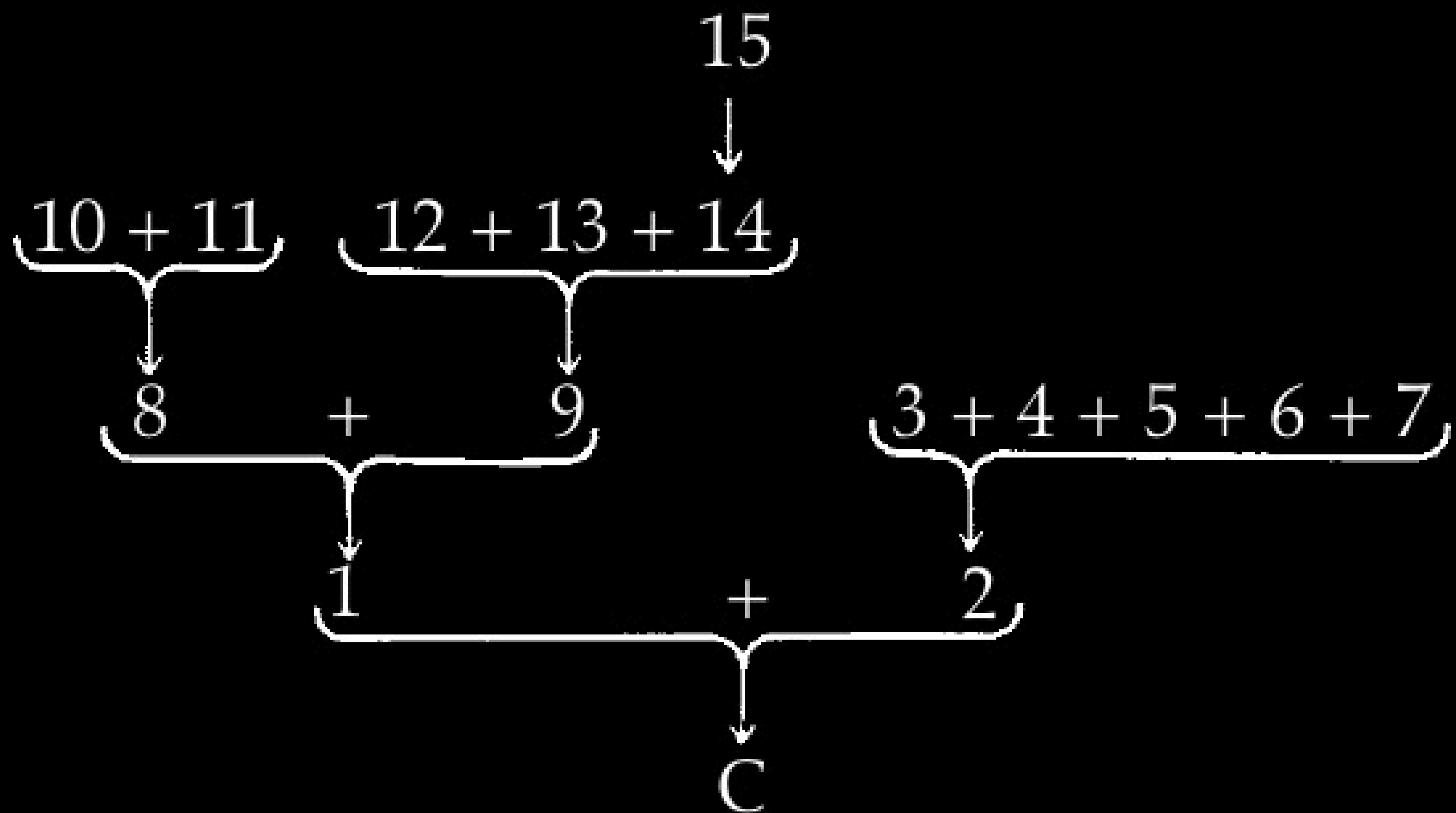
DOCUMENT

Coupling is the enemy of
change

DOCUMENT

This [coupling] makes change
more difficult

conclusion



Conclusions



**VALUE COMES
FROM
SOLVING
PROBLEMS**



GfK

GfK is the second largest market research company; they have lots of different data sources.



AUDI

We provided eight sessions of one week each for NLP, computer vision, and classical machine learning.



ZALANDO

Zalando has more than one hundred data scientists and is further down in data maturity than most companies.



VW

Since 2021, DSR trains all self-driving car computer vision employees for VW, the largest company in Germany

